Javier Vaca Sánchez
Marcos García Crespo
Daniel Hernández Utrilla



TIME SERIES COURSEWORK 2025-2026

INDEX

1. **Introduction and Data Description 1**
2. **Preliminary Analysis of the Time Series..... 1**
3. **Model Identification and Estimation (AR(1) and MA(1))..... 2**
4. **Residual Analysis and Model Validation..... 3**
5. **Model Selection and Overfitting 4**
6. **Forecasting and Model Comparison..... 6**
7. **Simulation and Spectral Analysis..... 8**
8. **Conclusion 9**
9. **Appendix 10**

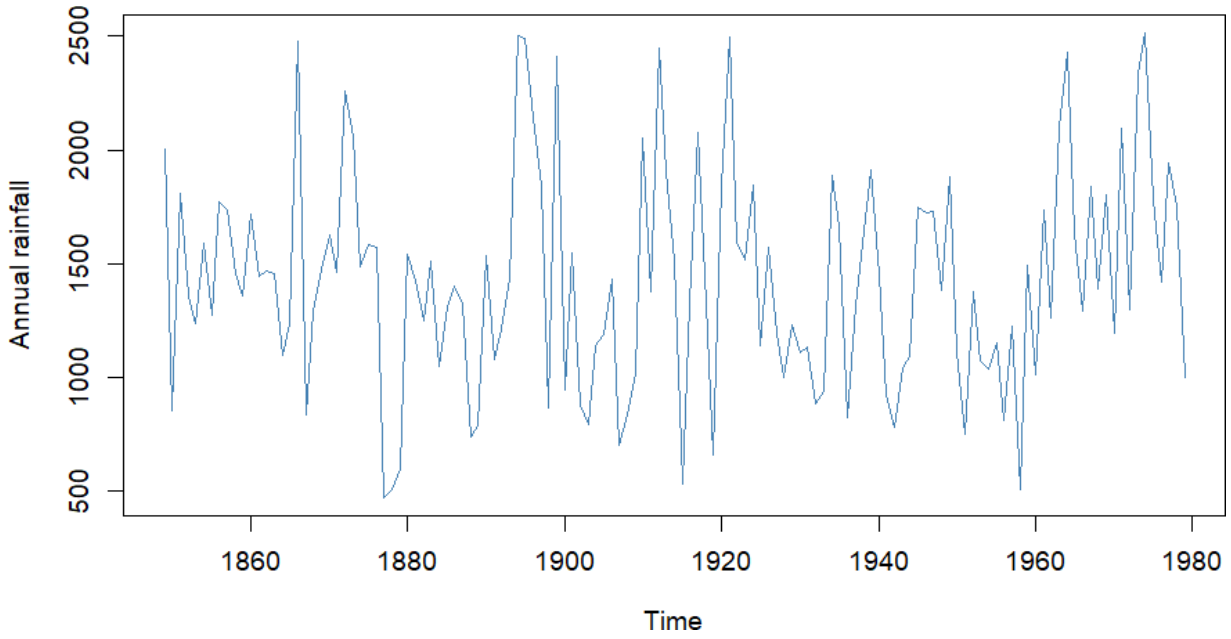
In this assignment we analyze a dataset named rain.txt, which was obtained from a database provided by the Universidad de Valladolid in Spain.

The dataset contains information on annual rainfall measurements recorded in the city of Fortaleza, Brazil, beginning in the year 1849.

For the analysis, we used the statistical software RStudio, together with the TSA package, which provides several tools for time series modelling and analysis.

Since the observations correspond to yearly data, the frequency parameter was not specified when creating the time series object. In this case, the default value of the ts function is 1, which is appropriate for annual data.

As a first step, we plot the time series in order to obtain an initial visual understanding of the behavior of the data that will be analyzed.

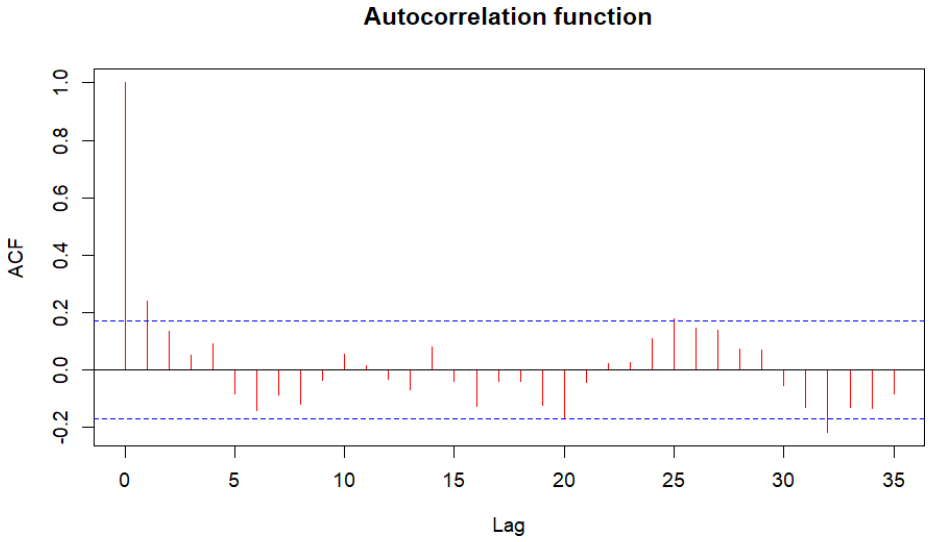


From the plot of the series, we observe that the rainfall values oscillate around approximately 1500 units (annual rainfall per m^2) and that this level remains relatively stable throughout the entire time period considered.

Concerning the variability of the observations, it appears to remain fairly constant across different segments of the series. Since neither the mean level nor the variance seem to change significantly over time, there is currently no clear evidence suggesting that the series is non-stationary.

Next, we compute the autocorrelation function (ACF) using a lag length of 35. The number of lags used in the autocorrelation analysis should typically fall between T/3 and T/4, where T represents the total number of observations. In our case, the dataset contains T = 131 observations, making a lag value of 35 appropriate.

1



The resulting plot shows that the first value, corresponding to lag 0, stands out because it is higher than the remaining ones. This behavior is expected, since the autocorrelation at lag 0 always takes the value 1, as it represents the correlation of the series with itself.

The blue confidence bands shown in the graph are useful for assessing the statistical significance of the autocorrelation values and for detecting possible dependence patterns within the time series.

Next, we proceed to estimate two different models for the data: an AR (1) model (autoregressive process of order one) and an MA(1) model (moving average process of order one). In our analysis, we denote the AR(1) model as m1, while the MA(1) model is referred to as m2.

```
m1<-arima(rain,order=c(1,0,0),method='ML')
m2<-arima(rain,order=c(0,0,1),method='ML')
```

The AR (1) fit gives us the following parameters. The formula is

$X_t = \omega + \varphi X_{t-1} + \varepsilon$, $\varepsilon \sim WN(0,\sigma^2)$

```
> m1

Call:
arima(x = rain, order = c(1, 0, 0), method = "ML")

Coefficients:
         ar1  intercept
      0.2421  1425.4842
s.e.  0.0851    53.5754

sigma^2 estimated as 217046:  log likelihood = -990.77,  aic = 1987.53
```

To estimate the parameters of the AR(1) model, we use the maximum likelihood method. The estimated value of the parameter φ is 0.2421, with a standard deviation of 0.0851. To determine whether this parameter is statistically significant, we construct a 95% confidence interval and check whether the value zero lies within that interval. For a confidence level of 95%, the critical value of the standard normal distribution is 1.96. Using this value together with the standard deviation of the estimator, we obtain the following interval for the parameter: (0.2421-0.0851*1.96, 0.2421+0.0851*1.96) = (0.075304, 0.408896).

2

Since the value 0 is not included in this interval, we reject the null hypothesis;

$$H_0: \varphi = 0$$

at the significance level $\alpha = 0.05$. This indicates that the parameter φ is statistically significant and should be considered in the model. For the MA(1) model, the process is defined by a constant term together with a white noise error term and the lagged value of that error term.

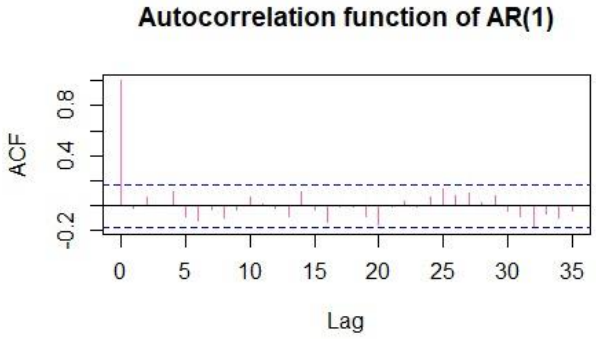
```
sigma^2 estimated as 21929:  log likelihood = -991.43,  aic = 1988.87
```

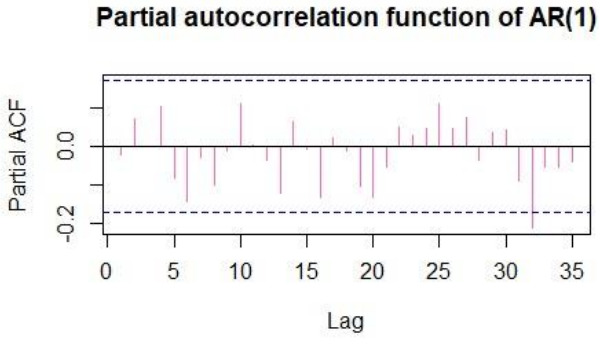
For the MA(1) model, the estimated value of the parameter θ is 0.2029, with a standard deviation of 0.0771.

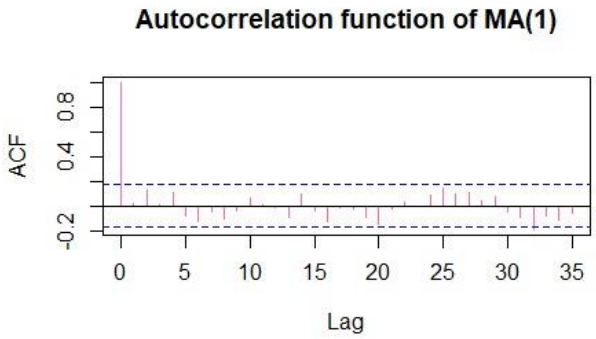
Using the same approach applied previously for the AR(1) model, we compute the 95% confidence interval for this parameter. The interval obtained is (0.2029-0.0771*1.96, 0.2029+0.0771*1.96) = (0.051784, 0.354016). Since the value 0 is not included within this interval, we reject the null hypothesis $H_0: \theta = 0$. This indicates that the parameter θ is statistically significant, and therefore it should be considered in our analysis.

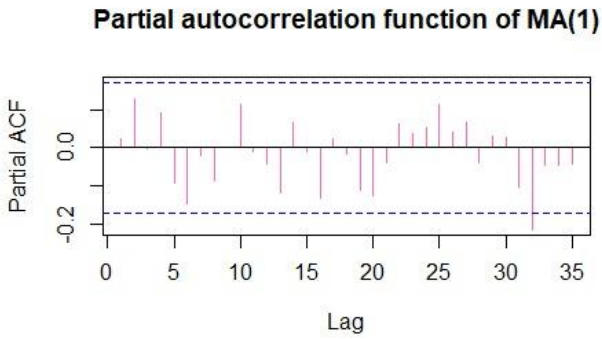
Once the parameters of both models have been estimated, the next step is to determine which model provides the best fit for the data. A common criterion for comparing competing models is the Akaike Information Criterion (AIC). In our case, the AIC values are 1987.53 for the AR(1) model and 1988.87 for the MA(1) model. Since the difference between these values is smaller than two units, it is generally considered that the two models have a very similar predictive performance. At this stage, both models remain possible candidates for describing the data.

To further assess their suitability, it is necessary to analyze the residuals of each model. For a model to be considered adequate, its residuals should behave like white noise, meaning that they have zero mean, constant variance, and are independent from one another. Therefore, after identifying the AR(1) and MA(1) models as potential candidates, we proceed to study their residuals in order to verify whether these conditions are satisfied. In particular, we examine whether the residuals show no significant autocorrelation, which would indicate that they follow a white noise process.









3

Each of the graphs provides useful information about the autocorrelation behavior of the residuals obtained from the AR(1) and MA(1) models.

The first graph presents the Autocorrelation Function (ACF) of the residuals from the AR(1) model. This function measures the correlation between residuals at different time lags. As can be observed, the autocorrelation values progressively decrease as the lag increases, which is consistent with the expected pattern of an AR(1) process.

The second graph displays the Partial Autocorrelation Function (PACF) of the residuals associated with the AR(1) model. The PACF measures the relationship between residuals at different lags while removing the influence of intermediate lags. In this case, the partial autocorrelation becomes negligible after the first lag, which is a typical feature of an AR(1) model.

The third graph corresponds to the Autocorrelation Function (ACF) of the residuals from the MA(1) model. As in the previous case, the ACF illustrates the correlation between residuals at different lags. The autocorrelation drops sharply after the first lag, which is a common characteristic of moving average processes of order one.

Finally, the fourth graph shows the Partial Autocorrelation Function (PACF) of the residuals corresponding to the MA(1) model. Similar to the second graph, the PACF measures the relationship between residuals at different lags while accounting for the influence of intermediate lags. In this case, the partial autocorrelation becomes insignificant after the first lag, which is a typical pattern for a moving average model of order one.

```
> Box.test(residuals(m1), lag = 20, type = "Ljung-Box", fitdf = 1)

        Box-Ljung test

data:  residuals(m1)
X-squared = 18.295, df = 19, p-value = 0.5028
```

The following output corresponds to the Ljung–Box test applied to the residuals of the model. The test statistic (X-squared) obtained is 18.295, while the number of degrees of freedom is 19. The associated p-value is 0.5028. Since this p-value is greater than the significance level of 0.05, we do not reject the null hypothesis of no autocorrelation in the residuals. Consequently, there is no statistical evidence suggesting the presence of autocorrelation, which supports the assumption that the residuals behave as white noise.

```
> Box.test(residuals(m2), lag = 20, type = "Ljung-Box", fitdf = 1)

        Box-Ljung test

data:  residuals(m2)
X-squared = 19.612, df = 19, p-value = 0.4183
```

Since the p-value obtained is 0.4183, which is higher than the commonly used significance level of 0.05, we cannot reject the null hypothesis that the residuals are not autocorrelated. This indicates that there is no sufficient statistical evidence to conclude that significant autocorrelation exists at the considered lags.

Another approach that can be used when selecting an appropriate model is overfitting the data. This method consists of introducing additional parameters into the model in order to determine whether a more complex specification provides a better description of the series.

In practice, we increase the number of parameters by one, either in the autoregressive component or in the moving average component of the model. Following this idea, we estimate three additional models: AR(2), which we denote as m3, MA(2), denoted as m4, and finally the ARMA(1,1) model, referred to as m5.

4

```
m3<-arima(rain,order=c(2,0,0),method="ML")
m4<-arima(rain,order=c(0,0,2),method="ML")
m5<-arima(rain,order=c(1,0,1),method="ML")
```

The AR (2) fit gives us the following parameters. The formula is
$X = \omega + \varphi 1X + \varphi 2X + \varepsilon$, $\varepsilon \sim WN\ (0,\sigma^2)$

```
sigma^2 estimated as 215780:  log likelihood = -990.39,  aic = 1988.78
```

In this case, the coefficient φ_2 is not statistically different from zero, since the value 0 lies within the corresponding confidence interval, which is (0.0769-0.0883*1.96, 0.0769+0.0883*1.96) = (-0.096168, 0.249968).

Because zero is included in this interval, we cannot reject the null hypothesis H_0: $\varphi_2 = 0$. This indicates that the parameter φ_2 is not statistically significant. As a result, the additional parameter introduced in the AR(2) model does not provide relevant information. Therefore, it is preferable to retain the AR(1) model, which is simpler and whose parameters are statistically significant.

MA(2): $X = \omega + \varepsilon - \theta 1\varepsilon_{t-1} - \theta 2\varepsilon_{t-2}$, $\varepsilon \sim WN\ (0,\sigma^2)$

```
> m4

Call:
arima(x = rain, order = c(0, 0, 2), method = "ML")

Coefficients:
         ma1     ma2  intercept
      0.2264  0.1044  1425.2248
s.e.  0.0892  0.0775    53.9469

sigma^2 estimated as 216332:  log likelihood = -990.55,  aic = 1989.11
```

In the MA(2) model, the coefficient θ_2 is not statistically different from zero. This is because the value 0 is included in the confidence interval, which is (0.1044-0.0775*1.96, 0.1044+0.0775*1.96) = (-0.0475, 0.2563).

Since zero lies within this interval, we cannot reject the null hypothesis H_0: $\theta_2 = 0$, indicating that the parameter θ_2 is not statistically significant. Consequently, the additional parameter introduced in the MA(2) model does not improve the model. For this reason, the simpler MA(1) model, whose parameters are statistically significant, provides a more suitable representation of the data.

ARMA(1,1): $X = \omega + \varphi X - \theta\varepsilon + \varepsilon$ $\varepsilon \sim N(0,\sigma^2)$

```
> m5

Call:
arima(x = rain, order = c(1, 0, 1), method = "ML")

Coefficients:
         ar1      ma1  intercept
      0.5034  -0.2788  1425.8626
s.e.  0.2518   0.2783    58.6631

sigma^2 estimated as 215807:  log likelihood = -990.4,  aic = 1988.79
```

In this case, the parameter θ is found to be not statistically significant. This is because the value 0 lies within the confidence interval, which is (-0.2788-0.2783*1.96, -0.2788+0.2783*1.96) = (-0.824268, 0.266668).

Since zero is included in this interval, we cannot reject the null hypothesis H_0: $\theta = 0$. Therefore, the parameter θ does not appear to contribute significantly to the model, indicating that the ARMA(1,1) specification is not suitable for describing the series. Consequently, we retain the AR(1) and MA(1) models, as these are the models in which all estimated parameters are statistically significant.

Once these two models have been selected, the next step is to analyze the correlation between the estimated

parameters. In particular, we study the relationship between the parameters ω and φ in the AR(1) model, and between ω and θ in the MA(1) model.

Having low correlation between the parameters of a model is desirable because it simplifies the interpretation of the results, increases the efficiency of the estimators, and can help reveal potential issues in the specification of the model. To evaluate the degree of correlation between the parameters, we compute the variance–covariance matrix associated with each model.

For the AR(1) model, the variance–covariance matrix is given by the following results.

```
                ar1   intercept
ar1       1.000000000 0.002744226
intercept 0.002744226 1.000000000
```

For the MA(1):

```
                ma1  intercept
ma1       1.00000000 0.00421732
intercept 0.00421732 1.00000000
```

In both models, the correlation between the estimated parameters is very small. This allows us to assume that the parameters are essentially uncorrelated, which is a desirable property since it simplifies the interpretation of the model and improves the reliability of the estimators.

The final specifications of the models are therefore the following:

AR(1):

$$X_t = 1425.4842 + 0.2421 X_{t-1} + \varepsilon_t$$

MA(1):

$$X_t = 1425.5435 + \varepsilon_t - 0.2029 \varepsilon_{t-1}$$

Having obtained two valid models capable of describing the series, the next step is to evaluate their predictive performance. In particular, we are interested in forecasting the values of the series for the next ten years, corresponding to the period 1980–1989.

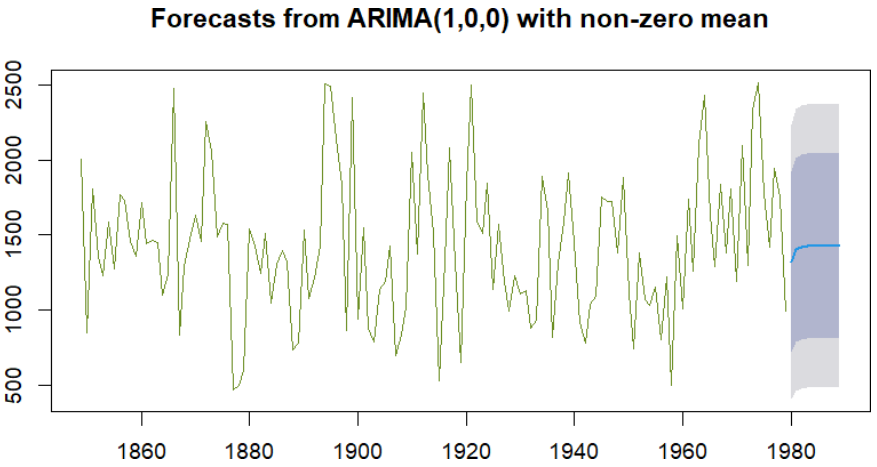
The following output presents the forecast results obtained from the AR(1) model.

```
> forecast(m1,h=10)
     Point Forecast    Lo 80    Hi 80    Lo 95    Hi 95
1980       1321.513 724.4606 1918.564 408.4004 2234.625
1981       1400.314 786.0162 2014.612 460.8265 2339.802
1982       1419.391 804.0972 2034.685 478.3804 2360.402
1983       1424.009 808.6571 2039.361 482.9094 2365.109
1984       1425.127 809.7717 2040.483 484.0222 2366.232
1985       1425.398 810.0421 2040.753 484.2925 2366.503
1986       1425.463 810.1076 2040.819 484.3580 2366.569
1987       1425.479 810.1235 2040.835 484.3739 2366.584
1988       1425.483 810.1273 2040.839 484.3777 2366.588
1989       1425.484 810.1283 2040.840 484.3786 2366.589
```

The forecast result for our MA(1) model:

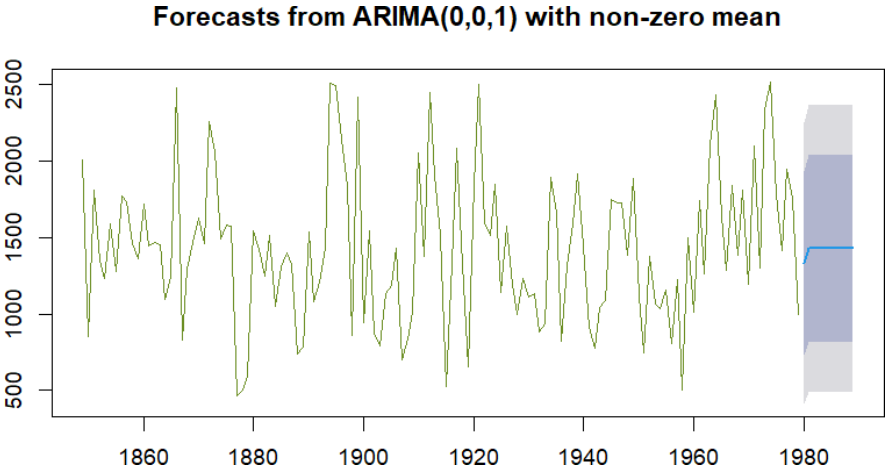
```
> forecast(m2,h=10)
     Point Forecast    Lo 80    Hi 80    Lo 95    Hi 95
1980       1329.321 729.1810 1929.461 411.4860 2247.156
1981       1425.543 813.1727 2037.914 489.0032 2362.084
1982       1425.543 813.1727 2037.914 489.0032 2362.084
1983       1425.543 813.1727 2037.914 489.0032 2362.084
1984       1425.543 813.1727 2037.914 489.0032 2362.084
1985       1425.543 813.1727 2037.914 489.0032 2362.084
1986       1425.543 813.1727 2037.914 489.0032 2362.084
1987       1425.543 813.1727 2037.914 489.0032 2362.084
1988       1425.543 813.1727 2037.914 489.0032 2362.084
1989       1425.543 813.1727 2037.914 489.0032 2362.084
```

The forecasted values are displayed in the second column, while the first column indicates the corresponding year. The remaining columns provide the lower and upper bounds of the confidence intervals, specifically for the 80% and 90% confidence levels associated with the predicted values. If we want to visualize both the original time series and the forecasts in a single graph, we simply need to include the plot command in the previous instructions.



If we consider the MA(1) model, we observe that the forecasts rapidly converge to a constant level, close to 1425.5435. This behaviour indicates that the influence of past shocks becomes negligible after a short period, causing the predicted values to stabilise around the mean of the series.

This suggests that the past error terms are relatively stable, and that the model does not identify strong patterns in the data beyond short-term fluctuations, which contrasts with the behaviour observed in the AR(1) model.



7

Finally, another criterion that can be used to compare models is their predictive performance. The model that provides more accurate forecasts can be considered the most appropriate. However, the evaluation of predictive ability depends on the specific forecasting objective.

For instance, if our goal is to predict the values of the series for the next ten years, we can reserve the last ten observations of the dataset as a validation sample. These observations are not used when estimating the models, but instead serve to evaluate how well each model performs in forecasting.

It is important to note that the same data should not be used both to estimate the model and to assess its predictive accuracy, since this could lead to misleading conclusions about the model's performance.

Following this approach, we compare the forecasting performance of the AR(1) and MA(1) models.

The sum of squared forecast errors was computed for both the AR(1) and MA(1) models. The values obtained were 2642062.3 for the AR(1) model and 2671282.9 for the MA(1) model. Since the error measure is smaller for the AR(1) model, this model provides more accurate forecasts and is therefore preferred.

It may also be useful to evaluate the models in terms of one-step-ahead predictions. In this case, the models are repeatedly re-estimated and the difference between the observed value and the corresponding prediction for the next period is calculated.

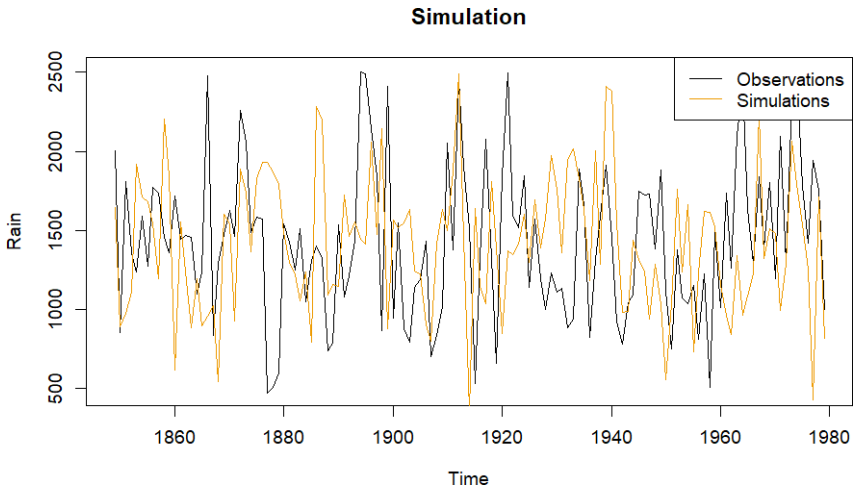
This procedure can be illustrated by applying it to the last five observations of the dataset.

In this case, the sum of squared forecast errors obtained for the two models are 597608.7 and 587651.7, respectively. Since the smaller value corresponds to the MA(1) model, this model produces more accurate one-step-ahead forecasts.

From these results, we can conclude that the choice of model depends on the forecasting horizon. When the objective is to obtain predictions further into the future, the AR(1) model tends to provide better performance. In contrast, when the focus is on short-term predictions, especially forecasts one period ahead, the MA(1) model yields slightly better results.

Because our main interest is in longer-term forecasts, we ultimately select the AR(1) model as the most suitable representation of the data.

To further examine the properties of this model, we conduct a simulation based on the estimated AR(1) process and this is what we obtain:



8

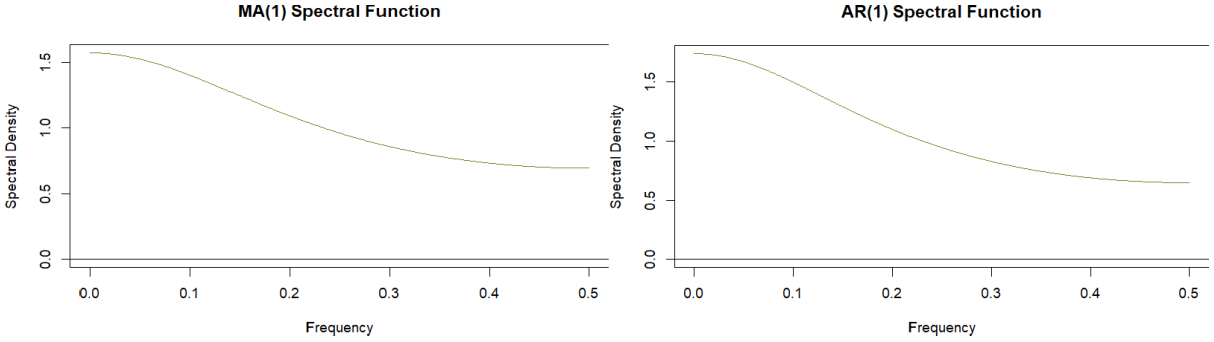
In this graph we can see the observations and the simulated data. We can conclude that the mean and variance are well estimated.

Next, we examine the frequencies represented in the spectral functions of the models. From the plot, we observe that the spectral curve decreases as the frequency increases, showing a pattern similar to an exponential decay.

This type of behaviour is typical of ARMA processes, particularly autoregressive models, where the autocorrelation between observations tends to diminish as the time lag becomes larger.

The progressive reduction in spectral density at higher frequencies may suggest the presence of noise in the series. In this context, the larger values of spectral density can be interpreted as representing the dominant signal frequencies, while the gradual decline may correspond to background noise in the data.

Overall, the spectral density plots indicate that both models display very similar spectral patterns.



In conclusion, the choice of the most appropriate model depends largely on the forecasting objective, particularly on whether the predictions are intended for the short term or the long term.

In our analysis, we first considered the AR(1) and MA(1) models as potential candidates. To determine which model better described the data, we analysed the residuals of each model by examining their autocorrelation and partial autocorrelation functions. In addition, we applied an overfitting approach, estimating the models AR(2), MA(2) and ARMA(1,1) in order to verify whether more complex specifications would improve the fit.

However, the additional parameters introduced in these models were not statistically significant, indicating that the extra complexity was unnecessary. Consequently, we retained the AR(1) and MA(1) models as the most appropriate representations of the series. After selecting these models, we also evaluated the correlation between their parameters using the variance–covariance matrix.

To further compare the models, we generated forecasts and analysed their predictive performance. The results showed that the AR(1) model performs better for long-term forecasts, whereas the MA(1) model provides slightly better results for one-step-ahead predictions.

9

APPENDIX

CODE IN R

```
library(TSA) library(forecast)

raindata<-read.table("rain.txt") rain<-ts(raindata,start=1849) View(raindata) nrow(rain) plot(rain, ylab="Annual rainfall",col="steelblue", xstart= c(1849))

acf(rain,lag.max=35,col="red",main="Autocorrelation function")

m1<-arima(rain,order=c(1,0,0),method='ML') m2<-arima(rain,order=c(0,0,1),method='ML')

c(0.2029-0.0771.96, 0.2029+0.0771.96) c(0.2421-0.0851.96, 0.2421+0.0851.96) par(mfrow=c(2,2))

acf(residuals(m1),lag=35,col="hotpink", main= "Autocorrelation function of AR(1)")
pacf(residuals(m1),lag=35,col="hotpink", main= "Partial autocorrelation function of AR(1)")
acf(residuals(m2),lag=35,col="hotpink", main= "Autocorrelation function of MA(1)")
pacf(residuals(m2),lag=35,col="hotpink", main= "Partial autocorrelation function of MA(1)")

par(mfrow=c(1,1))

Box.test(residuals(m1), lag = 20, type = "Ljung-Box", fitdf = 1)

Box.test(residuals(m2), lag = 20, type = "Ljung-Box", fitdf = 1)

m3<-arima(rain,order=c(2,0,0),method="ML") m4<-arima(rain,order=c(0,0,2),method="ML") m5<-arima(rain,order=c(1,0,1),method="ML")

c(0.0769-0.0881.96 , 0.0769+0.0881.96 ) c(0.1044-0.0771.96 , 0.1044+0.0771.96 ) c( -0.2788-0.2781.96,-0.2788+0.2781.96)

#variance-covariance matrix of the models var.m1<-m1$var.coef cor.m1<-var.m1/(sqrt(diag(var.m1))%*%t(sqrt(diag(var.m1)))) cor.m1

var.m2<-m2$var.coef cor.m2<-var.m2/(sqrt(diag(var.m2))%*%t(sqrt(diag(var.m2)))) cor.m2

forecast(m1,h=10) forecast(m2,h=10)

dd<-data[1:121,1] ddts<-ts(dd,start=1849)

m1p<-arima(ddts,order=c(1,0,0),method="ML") m2p<-arima(ddts,order=c(0,0,1),method="ML") par1<-forecast(m1p,h=5) pma1<-forecast(m2p,h=5) difar1<-0 difma<-0 for(i in 1:5){ difar1<-difar1+(data[121+i,1]-par1$mean[i])^2 difma1<-difma1+(data[121+i,1]-pma1$mean[i])^2 }

difar1<-0 difma<-0 for(i in 1:5){ dd<-datos[1:125+i,1] ddts<-ts(dd,start=1849) m1p<-arima(ddts,order=c(1,0,0),method="ML") m2p<-arima(ddts,order=c(0,0,1),method="ML") par1<-forecast(m1p,h=1) pma1<-forecast(m2p,h=1) difar1<-difar1+(data[121+i,1]-par1$mean[i])^2 difma1<-difma1+(data[121+i,1]-pma1$mean[i])^2 }
```


10

```
ar1.model = list(ar=0.2421) mu<-1425.4842 n<-131 sd<-sqrt(217046)

set.seed(123) ar1.sim2 = mu + arima.sim(model=ar1.model, n=131, innov=rnorm(n=131,mean=0,sd=sd))

plot(rain,ylab="Rain",main="Simulation") tss<-ts(ar1.sim2,start=1849)
points(tss,col="rosybrown3",lty=1,type="l")

legend("topright", legend=c("Observations", "Simulations"), col=c("black", "rosybrown3"), lty=c(1,1),
title="data:")

ARMAspec(model=list(ar=m1$coef[1]), main="AR(1) Spectral Function", col="olivedrab")
ARMAspec(model=list(ar=m2$coef[1]), main="MA(1) Spectral Function", col="olivedrab")
```


11